

Closed-Form Time-of-Flight Anchoring for Real-Time Metric Depth on an Embedded Robot

Leo Xie

Dept. of Mechanical and Mechatronics Engineering
University of Waterloo
Waterloo, Canada
ly4xie@uwaterloo.ca

Eva Xie

Exo Tech Society
Toronto, Canada
eva@exotechociety.org

Abstract—Autonomous robots too small to carry a scanning LiDAR still need dense metric depth to avoid obstacles and build maps. A multizone direct time-of-flight (dToF) sensor gives absolute range at low resolution, while a camera gives high-resolution structure with no absolute scale. Existing methods learn the mapping between the two end-to-end, but report results on desktop-class GPUs, which leaves open whether depth fusion is practical on a robot’s own embedded compute budget. We address this question with a system that aligns camera-predicted depth to dToF zones by fitting just two parameters, a scale and a shift, in a single weighted least-squares solve that requires no training. Evaluated on the ZJU-L5 benchmark against dense independent ground truth, with nothing tuned on the dataset, the fit on a shared public backbone reaches 0.902 threshold accuracy and 0.094 absolute relative error with no learned fusion parameters, ahead of two baselines trained on the benchmark. The deployed variant swaps that backbone for a distilled student of 3.66 million parameters, trading benchmark accuracy for speed, and on an NVIDIA Jetson AGX Orin it runs 3.1 times faster than the DEPTHOR-Small network at matched resolution. We also characterize where the approach breaks down, and discuss limitations and future work.

Index Terms—Robotic perception and localization, perception and sensing in autonomous systems, sensor fusion, direct time-of-flight sensors, monocular depth estimation, embedded systems, point clouds

I. INTRODUCTION

Autonomous robots build maps and avoid obstacles from a metric point cloud, and on most platforms a scanning LiDAR produces it. Small robots often cannot fit one: a spinning LiDAR needs vertical clearance and power that a compact platform may not have, and a stereo rig needs a baseline. Nearly all of them, however, already carry a camera, which raises the question this paper addresses: whether a camera and a solid-state range sensor together can supply what robotic perception and localization need within the space a compact platform can provide.

A multizone direct time-of-flight (dToF) sensor is an ideal candidate: it is solid-state and inexpensive, and it returns true absolute range, but current devices report only 8×8 to 48×32 zones. A monocular depth network gives far more detail, but its output is not metric, because a single image fixes depth only up to an unknown scale, and for affine-invariant models up to an unknown shift as well. The camera supplies structure without absolute scale, and the dToF absolute scale

without structure. Recent systems combine them with learned fusion [1], [2] or completion [3], [4] networks, which lead on accuracy but are timed on desktop GPUs. Whether the same depth is achievable on the processor a robot carries, which delivers far less throughput inside a fixed power and thermal budget, is left open.

That gap leaves three questions open. Does metric alignment require a learned fusion module at all, or do the range measurements already overdetermine it? Can a pipeline built on that answer hold real-time rates under an embedded memory and power budget? Can it report where its own output is untrustworthy, so that a robot acting on the depth map separates measurement from inference?

Our design follows from counting unknowns against measurements. Converting an affine-invariant prediction to metric depth requires only two numbers for the whole image, a scale and a shift, while a 32×32 dToF tile supplies up to 1024 range measurements. That is roughly five hundred measurements for every unknown, far more than are needed to determine both. Metric alignment is therefore an estimation problem rather than a learning problem: a weighted least-squares fit solves it in closed form in microseconds, and the covariance of that fit supplies an uncertainty estimate directly. We keep learned components only for the two tasks that need them, relative structure and local correction, and neither network receives a dToF measurement as input. That answer to the first question brings a fourth problem with it, since the depth an affine fit can express has a structural limit.

Dropping the learned fusion module also shrinks dense computation to two small convolutional networks. The same analytic fit, run on that method’s own backbone, lands within 4 to 8% of DEPTHOR-Small [3] on relative error. That accuracy figure comes from the shared public backbone rather than the distilled one the deployed system runs, a distinction Section V-F keeps explicit. The analytic stage also accepts a different calibrated sensor without retraining, since a new zone layout changes only the rows of a linear system. We make the following contributions:

- A closed-form anchoring stage driven by active range measurements on a known angular grid, which admits range-dependent weighting and recovers scale without egomotion, an inertial unit, or texture.

- A residual refiner that corrects the affine parameters themselves, so it composes with the analytic stage, and a zero-initialized output layer that leaves the untrained system as the identity.
- An empirical finding about supervising that refiner: a scattered hold-out yields nothing, while a contiguous held-out island improves extrapolation by 40%.
- A characterization of where affine anchoring breaks down: expressible depth is capped at $1/b$, and the analytic variance term inherits that cap.
- An embedded implementation that runs $3.1\times$ faster than DEPTHOR-Small’s network.

II. RELATED WORK

A. Lightweight dToF and RGB fusion

DELTAR [1] introduced depth estimation from an 8×8 dToF sensor and an RGB image, representing each zone as a range distribution and fusing with attention. CFPNet [2] propagates information beyond the zone footprint. DEPTHOR [3] recasts the task as completion around a frozen monocular prior with simulated dToF noise, while DEPTHOR++ [4] adds anomaly rejection, SelfToF [5] removes the ground-truth requirement, and Prompt Depth Anything [6] conditions a depth foundation model on sparse metric input. General depth completion [7], [8] is accurate but not built around multizone dToF geometry. All of these place metric fusion inside a learned module and report desktop-GPU results, so what such a module costs on a robot’s own limited processor is left open. Table I summarizes where each method performs metric fusion and what its anchors require.

B. Metric alignment of monocular priors

Monocular models such as Depth Anything V2 (DAv2) [9], Metric3D V2 [10], and UniDepth [11] supply strong relative geometry, but affine-invariant predictors must still be metrically aligned at inference. VI-Depth [12] performs that alignment by fitting a global scale and shift to sparse visual-inertial landmarks, then regressing a dense scale map to correct what the global fit leaves behind. Its anchors are triangulated, so they require camera motion, an inertial unit, and scene texture, and they fall wherever image features happen to land rather than on a known grid. It is also evaluated on a desktop GPU. Metric alignment from a solid-state range sensor, and its cost and speed on embedded hardware, therefore remain open.

III. SYSTEM OVERVIEW

Our sensor module pairs an Arducam B0179 IMX219 camera, fitted with a 2.5 mm M12 fisheye lens and captured at 1640×1232 , with an AMS TMF8829 dToF tile operating in its 32×32 focal-plane mode. An ESP32-C6 microcontroller polls the tile over I2C, timestamps each subframe in hardware, and streams framed binary packets over USB. All computation runs on an NVIDIA Jetson AGX Orin in its maximum performance mode. The lens centre sits 20.2 mm above the dToF centre on a common vertical axis, an offset carried in

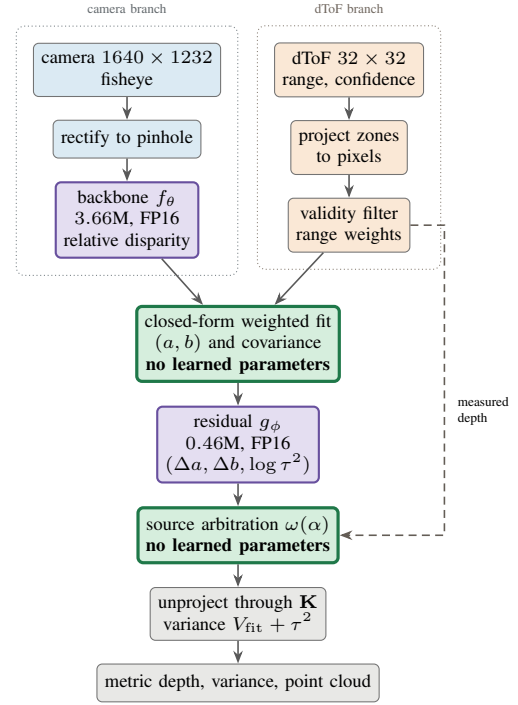


Fig. 1. Purple marks the two trained networks, which handle relative structure and local correction and never receive a dToF measurement. Green marks the two stages that establish and allocate metric scale, neither of which has learned parameters. The dToF is consumed twice, as anchors for the fit and as depth near a measurement.

the calibrated extrinsic. Fisheye intrinsics were calibrated to a reprojection error of 0.54 px, and the dToF cone measures $73.5^\circ\times 60.5^\circ$ against the manufacturer’s nominal 80° diagonal, its projected footprint covering 12.4% of the frame.

Fig. 1 shows the system architecture, which runs in eight stages across two branches. The camera branch rectifies the fisheye frame to a pinhole image and passes it to a distilled 3.66M-parameter backbone predicting relative inverse depth. The dToF branch converts each zone to a three-dimensional point from its measured range and calibrated ray, projects that point into the image, and rejects returns that fail the validity check or fall outside the frame.

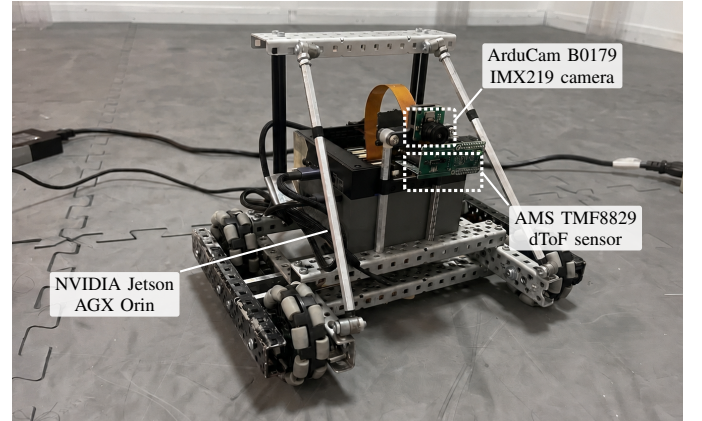


Fig. 2. The sensor module on the test platform.

TABLE I
WHERE METRIC FUSION HAPPENS, AND WHAT THE ANCHORS REQUIRE

Method	RGB model	Metric fusion	Local refinement	Needs motion	Platform
DELTAR [1]	EfficientNet	learned attention	learned head	no	desktop GPU
CFPNet [2]	encoder decoder	learned propagation	learned	no	desktop GPU
DEPTHOR [3]	frozen DAv2	learned completion	learned completion	no	desktop GPU
DEPTHOR++ [4]	frozen DAv2	filter, then learned	learned completion	no	desktop GPU
VI-Depth [12]	MiDaS, large	closed form (visual-inertial)	learned dense scale map	yes	desktop GPU
Ours	distilled 3.66M	closed form (dToF zones)	residual on (a, b) , 0.46M	no	Jetson AGX Orin

The branches meet at the anchoring stage, where each surviving zone’s pixel samples the backbone prediction and contributes one row of a linear system, whose closed-form solution is the scale and shift that convert the whole disparity field to metres. A 0.46M-parameter residual refiner then corrects those two parameters per pixel, where a single global transform does not hold.

An arbitration rule sets how much of each published value comes from measurement rather than inference, and every pixel is unprojected through the camera intrinsics $\mathbf{K}$ into a metric point cloud carrying a per-pixel variance. Only the backbone and the refiner hold learned parameters, and neither performs the metric fusion.

IV. METHOD

The pipeline turns one affine-invariant disparity field and a few hundred range measurements into dense metric depth, confining metric information to a single two-parameter fit and leaving to the learned components only what that fit cannot do. The subsections below take each stage in the order data moves through it.

One preliminary applies throughout. Because monocular depth networks are trained on rectilinear imagery, each fisheye frame is remapped to a pinhole image through an offline-calibrated Kannala-Brandt model, using a precomputed lookup table so the runtime cost is a memory-bound copy and every stage downstream works under that one intrinsic matrix.

A. Relative depth backbone

The camera branch supplies the dense structure that every later stage rescales. The rectified image is processed by the backbone,

$$\hat{\delta}_t = f_\theta(I_t), \quad (1)$$

where f_θ is the backbone, I_t the rectified frame at time t , and $\hat{\delta}_t$ the relative inverse depth it predicts. The model has 3.66M parameters, a MobileNetV3-Large encoder with features at strides 4, 8, 16, and 32, and a lightweight decoder of four fusion blocks emitting one disparity channel at 384×288 . It uses only regular convolutions, bilinear resizing, and elementwise operations at fixed tensor shapes, with no attention over image tokens and no coordinate-indexed gather or scatter. Those are the memory-bound patterns that dominate attention-based fusion runtime and quantize poorly, and avoiding them is what permits compilation to efficient TensorRT kernels.

The backbone is distilled from Depth Anything V2 [9]. Teacher and student are both affine-invariant, so the student

is first aligned to the teacher by the same scale and shift fit used at inference, after which training minimizes a trimmed absolute error and a multi-scale gradient term,

$$\mathcal{L}_{\text{distill}} = \text{trimL1}(\mathcal{A}(\hat{\delta}^s), \hat{\delta}^t) + \lambda_g \sum_k \left\| \nabla_k \mathcal{A}(\hat{\delta}^s) - \nabla_k \hat{\delta}^t \right\|_1, \quad (2)$$

where trimL1 discards the worst 20% of residuals before averaging, which stops the student reproducing teacher failures on glass and specular surfaces. Because supervision is the teacher’s output rather than measured depth, distillation needs only rectified images, so the student can be trained on frames from the deployment camera alone. That is a deliberate trade, the student becomes accurate on this lens and loses accuracy on images from any other camera, which Section V-F quantifies on a public benchmark.

B. Zone projection

The backbone output carries no units. The dToF supplies them, but only once each zone is matched to the pixels it actually observes. For each valid zone i with measured range ρ_i and calibrated ray $\mathbf{r}_i$, the corresponding point is transformed into the camera frame and projected,

$$\mathbf{P}_i^c = \mathbf{T}_{c \leftarrow d} \rho_i \mathbf{r}_i, \quad \mathbf{p}_i = \pi(\mathbf{K}, \mathbf{P}_i^c). \quad (3)$$

The metric target is the camera-frame axial depth $Z_i = P_{i,z}^c$, not the slant range ρ_i , since using the latter introduces an error that grows with angle from the optical axis. Because the range is measured, the parallax from the 20.2 mm sensor separation is compensated exactly. The projection must be exact in both grid orientation and field-of-view assignment: an error in either places every anchor on the wrong pixel while leaving the pipeline superficially functional. Anchors that fall outside the image, have non-positive axial depth, or fail the validity check are rejected, leaving

$$\mathcal{A}_t = \{(\hat{\delta}_i, s_i, w_i, \mathbf{p}_i)\}, \quad s_i = 1/Z_i. \quad (4)$$

in which each entry of the anchor set $\mathcal{A}_t$ pairs the backbone prediction at pixel $\mathbf{p}_i$ with the inverse depth measured there.

C. Closed-form metric anchoring

Each anchor now sits on a known pixel, so it can be paired with the backbone’s prediction there, and the two quantities become comparable. Affine-invariant inverse depth relates to metric inverse depth by $s_i = a\hat{\delta}_i + b + \epsilon_i$, so with $\mathbf{X} = [\hat{\delta} \ 1]$ and $\mathbf{W} = \text{diag}(w_i)$ the weighted least-squares estimate is

$$\hat{\beta} = [\hat{a} \ \hat{b}]^\top = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{s}, \quad (5)$$

where $\mathbf{X}$ has one row per anchor but two columns, so $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ stays 2×2 however many anchors there are. Building it needs five running sums over the anchors, $\sum w_i$, $\sum w_i \hat{\delta}_i$, $\sum w_i \hat{\delta}_i^2$, $\sum w_i s_i$ and $\sum w_i \hat{\delta}_i s_i$, and the solution then follows from inverting a two-by-two matrix. One robust reweighting pass limits the influence of multipath returns. The fit is identifiable when the anchors span nonzero weighted variance in predicted disparity; a scene at uniform depth is the degenerate case, for which the system falls back to a scale-only estimate. The anchored depth is

$$D_t^0(\mathbf{p}) = \left(\hat{a} \hat{\delta}_t(\mathbf{p}) + \hat{b} \right)^{-1}. \quad (6)$$

Changing the zone count or geometry changes the rows of $\mathbf{X}$ and nothing else, which is what makes this stage layout independent.

Least squares in inverse-depth space implicitly favours near anchors, since a point at 0.3 m enters the residuals at 3.3 m^{-1} against 0.4 m^{-1} for one at 2.4 m, and with uniform weights the far field is under-read by roughly 30%. Propagating relative depth error predicts a correction of $w_i \propto z_i^2$, but measurement refutes it: that weighting overshoots into a far-field over-read because the affine model is misspecified. We use $w_i \propto z_i$ and record the disagreement, since it is the clearest evidence that what limits the fit is the affine model rather than the noise model.

Equation (6) also sets a limit that every result in this paper inherits. As $\hat{\delta}_t \rightarrow 0$, meaning an arbitrarily distant surface, $D_t^0 \rightarrow 1/\hat{b}$. The shift term therefore sets a hard ceiling on the deepest value the model can represent, and $\hat{b}$ is fitted entirely from anchors inside the dToF cone. No part of the fit relates in-cone measurements to out-of-cone range, so structure beyond $1/\hat{b}$ cannot be reported however clearly the image shows it. Section V-D quantifies where this binds.

D. Metric residual refiner

The fit above applies one pair of parameters to every pixel in the frame. That is what makes scale recoverable from a few hundred anchors, and also what leaves every error varying across the image uncorrected. One global affine map cannot correct local error in the monocular prediction, which concentrates at depth discontinuities, on thin structures, and far from any anchor. The refiner g_ϕ is a 0.46M U-Net taking the rectified image, the log-normalized anchored depth, the sparse projected anchor depths, and a binary anchor mask, and emitting

$$(\Delta a, \Delta b, \log \tau^2) = g_\phi(I_t, D_t^0, S_t, M_t), \quad (7)$$

which gives the refined depth

$$D_t(\mathbf{p}) = \left[(\hat{a} + \Delta a(\mathbf{p})) \hat{\delta}_t(\mathbf{p}) + \hat{b} + \Delta b(\mathbf{p}) \right]^{-1}. \quad (8)$$

Correcting the affine parameters rather than the depth itself keeps the learned stage in the estimator's own parameterization, so the two compose rather than stack, since the refiner's correction lives in the same parameter space as the fit instead of layering a second depth map on top. Setting $\Delta a = \Delta b = 0$

recovers D_t^0 exactly, and the output layer is zero-initialized, so an untrained refiner is the identity and the deployed system has a deterministic analytic fallback.

Training withholds a subset of each frame's valid zones and splats them as sparse targets, so the refiner is scored where a real measurement exists but it received none. Because withheld anchors constrain only the cone, a multi-scale term matches $\nabla \log D_t$ to $\nabla \log D_t^0$ elsewhere, tying the refined field to the backbone's geometry where no measurement exists. The term is zero when the refiner is the identity, so a converged identity solution is unpenalized. Which zones are withheld determines what the refiner learns, an effect Section V-B quantifies.

E. Source arbitration

The camera branch now gives a metric value at every pixel, while the dToF still holds direct measurements over part of the frame, so the same surface is described twice. The two estimators have complementary weaknesses. The dToF measures true distance but covers only 12.4% of the frame, while the refined camera path covers every pixel and inherits any error in the fit. Neither is right everywhere, so the pipeline needs a rule for which to believe at each pixel, and the natural variable is angular distance to the nearest valid measurement. We publish

$$\tilde{D}_t(\mathbf{p}) = \omega(\alpha_{\mathbf{p}}) D_t^{\text{dToF}}(\mathbf{p}) + (1 - \omega(\alpha_{\mathbf{p}})) D_t(\mathbf{p}), \quad (9)$$

where $\alpha_{\mathbf{p}}$ is the angle to the nearest anchor and ω descends from one to zero between 2° and 5° along a smoothstep profile. Using angle rather than pixel distance keeps the rule unchanged across zone layouts. The stage also improves as the sensor improves, since more zones widen the region where a measurement is nearby, and ω doubles as a per-pixel record of how much of a depth value came from measurement.

F. Uncertainty

The fit of Section IV-C returns more than two numbers, its normal system also describes how well they are determined. The covariance of the fitted parameters follows from that same weighted normal system, $\Sigma_\beta = \hat{\sigma}^2(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$. First-order propagation through the inverse map, with $\mathbf{j}(\mathbf{p}) = [\hat{\delta}_t(\mathbf{p}), 1]^\top$, gives

$$V_{\text{fit}}(\mathbf{p}) \approx D_t^4(\mathbf{p}) \mathbf{j}^\top(\mathbf{p}) \Sigma_\beta \mathbf{j}(\mathbf{p}), \quad (10)$$

which captures uncertainty in the fitted scale and shift but not all predictive uncertainty in the monocular model. The refiner predicts the remainder τ^2 under a Gaussian negative log likelihood, penalizing both overconfidence and excessive width, and the published variance is $V_t = V_{\text{fit}} + \tau^2$. The analytic term inherits the ceiling of Equation (6), since $V_{\text{fit}} \propto D_t^4$ and D_t is capped at $1/\hat{b}$. However, the published interval is not bounded by that cap because τ^2 dominates V_t in the deployed configuration. Section V-E reports what does.

G. Output

One step remains, turning the arbitrated field into the point cloud a navigation and mapping stack expects. Every pixel is unprojected through $\mathbf{K}$, inheriting the variance of its source pixel, so the published product is a metric point cloud rather than a depth image alone. At the deployed stride it carries roughly 125k points per frame against the 1024 zones the sensor measures directly, so the dToF supplies scale and the camera supplies structure. Fig. 3 shows the result on a moving platform.

V. EXPERIMENTAL RESULTS

Every timing figure was taken on the deployed hardware with the Jetson in maximum performance mode, both networks running as TensorRT FP16 engines with arbitration, unprojection, and variance propagation on the GPU. Accuracy for the larger public backbones was computed offline on the same code path. The backbone was distilled from 2000 rectified frames collected through the world environment path, reaching a Pearson correlation of $\rho = 0.996$ with the teacher’s disparity. An earlier distillation set overlapped with our own evaluation logs, so we re-distilled on 1800 frames with the remaining 200 held out, and every own-hardware figure below uses that clean configuration.

Because the sensor gives no dense ground truth, accuracy on our own hardware is measured by withholding zones and scoring where a measurement exists but the pipeline received none. The *random pattern* protocol scatters those zones, so a surviving measurement sits about one zone pitch away, 2.3° horizontally and 1.9° vertically, which probes interpolation. The *center pattern* protocol keeps a contiguous central island and withholds everything outside it, probing extrapolation to about 25° from the optical axis. Both are genuine hold-outs: fitting and scoring on the same anchors would let a nearest-zone lookup score exactly 0.000 m.

Four sets of measurements follow: 1234 logged camera-dToF pairs, 600 driving frames with anchors confined to a central island, 300 logs for the regularization sweep of Section V-D, and a tape-measured reference. That reference was collected in two sessions, seven surveyed markers in the first and 15 points across the two combined, spanning 0.33 to 4.03 m with five repeat captures. Four of the 15 points lie outside the dToF cone. We report threshold accuracy δ_1 , absolute relative error (AbsRel), root mean square error (RMSE), mean absolute error (MAE), and median absolute error, with coverage and signed bias. Depth error here is heavy-tailed, so wherever the shape of that tail is at issue we give a mean and a median together. In the tables, $\downarrow$ and $\uparrow$ mark direction and $\rightarrow 0$ or $\rightarrow 1$ a target.

Our external benchmark is ZJU-L5 [1], which supplies 8×8 VL53L5CX measurements with dense RealSense ground truth over a 527-pair test split, and is our only public benchmark with ground truth independent of the sensor being fused. It is zero-shot for the backbone, which saw none of this dataset, its train split, or its camera, subject to two qualifications: a scalar ridge strength was selected on the train split, reported in

TABLE II
LATENCY AT 480×640 ON THE JETSON AGX ORIN

Configuration	latency $\downarrow$	rate $\uparrow$
DEPTHOR-Small [3]	79.4 ms	12.6 Hz
DEPTHOR-Large [3]	183.8 ms	5.4 Hz
Ours, arbitration and ROI off	20.0 ms	50.1 Hz
Ours, deployed configuration	25.8 ms	38.7 Hz

Section V-D, and neither our zero-shot claim nor DEPTHOR’s covers backbone pretraining, since both inherit whatever the shared DAv2 weights saw. Running the released DEPTHOR-Small weights through our harness on the same Jetson reproduces its published metrics to within 0.002 to 0.008 on every metric, which is what licenses the comparisons below.

Six results follow: throughput and latency, accuracy against withheld zones, validation against an independent tape reference, the structural limit bounding every accuracy number here, whether the predicted uncertainty can be trusted, and how the system compares against published methods.

A. Throughput and latency

Measured on the robot in its default configuration, including ROS transport and rectification, the node publishes at 9.4 Hz. The same perception core measured offline runs roughly twice as fast, since an isolated benchmark has the Orin’s shared LPDDR5 to itself. The dToF delivers subframes at 16 Hz but complete 32×32 maps assemble at 8.3 Hz, so the node publishes faster than the sensor refreshes its anchors and the 9.4 Hz rate is set by compute rather than by waiting on the sensor.

Rate alone does not characterize a pipeline feeding a controller. The sensor supplies frames faster than the pipeline consumes them, so an unbounded queue would work through a backlog of obsolete frames at unchanged throughput. The input queue is therefore bounded, and depth maps arrive a median of 128 ms old against the 106 ms publish period that 9.4 Hz implies, a distinction no rate measurement can detect.

Against a learned baseline, Table II reports network-only latency at matched resolution, batch one, CUDA-synchronized, with the GPU otherwise idle. We ran DEPTHOR’s released weights on our own Orin, so the comparison measures their model rather than our reimplementations of it, and their paper names inference speed as the method’s main limitation while reporting no timing. In the deployed configuration our complete pipeline is $3.1\times$ faster than DEPTHOR-Small’s network alone, rising to $4.0\times$ with the two runtime-optional stages, source arbitration and the region-of-interest (ROI) crop, switched off. Since the table compares a pipeline against a network, the deployed 9.4 Hz at full resolution is the figure to use when comparing whole systems.

B. Accuracy on our own hardware

Speed is worth nothing if the depth is wrong, so the rest of this section is accuracy. Table III scores three configurations against withheld zones, under both protocols and by angle from the optical axis. The analytic output is the reference

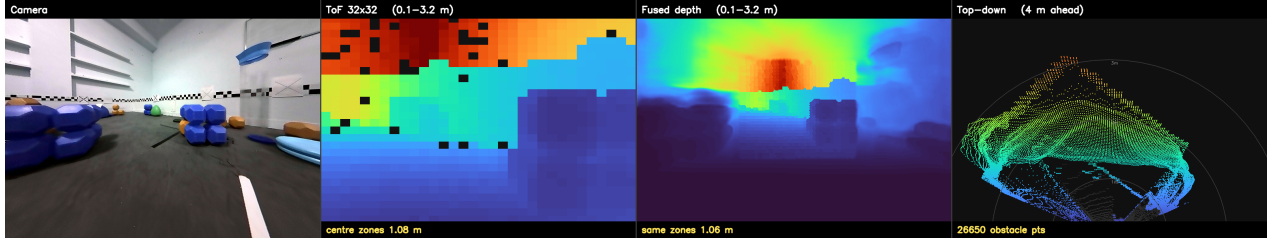


Fig. 3. The deployed system on a moving platform, recorded from the published topics, with both depth panels on a common 0.1 to 3.2 m scale. Left to right: the rectified camera frame; the 32×32 dToF returns, covering 12.4% of the field of view; the fused metric depth; and a top-down view within 4 m ahead, where 26,650 points fall in the obstacle band out of roughly 125k in the full cloud. Over the centre zones the sensor reads 1.08 m and the fused output 1.06 m at those same zones, so anchoring preserves the measurement where one exists. The second panel is all the sensor measures directly; the third is what the pipeline publishes from it on the robot’s own processor.

TABLE III
MEDIAN ABSOLUTE ERROR ON 1234 LOGGED PAIRS, IN m

Method	protocol		center, by angle from optical axis				
	rand.↓	cen.↓	0–3°	3–6°	6–10°	10–15°	15–30°
nearest-zone dToF	0.018	0.109	0.027	0.063	0.104	0.163	0.204
refiner, island	0.050	0.033	0.033	0.028	0.031	0.032	0.036
arbitration	0.014	0.032	0.026	0.027	0.031	0.032	0.036

throughout, since a zero-initialized refiner reduces to it exactly, scoring 0.055 m on both.

On the random pattern protocol the raw sensor beats the refiner outright, 0.018 m against 0.050 m, since copying a real measurement beats inferring one, so no accuracy claim for the learned path can rest on interpolation. Across the angular bands it degrades by roughly $8\times$ while the camera path stays approximately flat, and the two cross at about 3° . The cone covers only 12.4% of the frame, so the outer bands are the operating regime, and that crossover is what Equation (9) exploits.

What changed the refiner was the geometry of its supervision rather than its capacity. A scattered 25% hold-out leaves a real measurement one zone from almost every supervision target, so the network learns only interpolation and on extrapolation performs no better than no network at all, 0.066 m against the analytic 0.064 m on the split where both variants were compared. Supervising instead on a held-out outer island, one changed training argument with the same architecture and data, takes extrapolation to 0.033 m, and a mixed hold-out lands between the two. The arbitration rule of Section IV-E then recovers the better of the two everywhere: it defers to the sensor where a raw reading is best and reaches 0.032 m in extrapolation, matching or beating either source alone in every angular band.

Arbitration also holds up under motion. We compare two variants over 600 driving frames, identical except for arbitration and sharing one backbone and residual pass per frame. Over 386,681 scored zones outside the anchor island, MAE falls from 0.264 to 0.247 m and median frame-to-frame change improves from 0.0106 to 0.0100 m. The gain concentrates where the design predicts, with median error at $15\text{--}30^\circ$ from the island falling from 0.090 to 0.069 m and the $10\text{--}15^\circ$ band reaching 0.008 m. Beyond 30° the two variants are numerically identical, since no anchor is close enough to blend

toward. Only the within-run gap is controlled here, as route and speed differ between sessions.

C. Validation against an independent reference

Every result above is scored against the same dToF the pipeline anchors to, which establishes self-consistency rather than correctness: anchors and evaluation targets are the same points, so an error common to both cancels. Over the seven markers of the first session the deployed system reaches a median error of 0.044 m, MAE 0.192 m, RMSE 0.321 m, and a maximum of 0.774 m.

Five of the seven markers agree with tape to within 4 cm, and the remaining two carry the gap between median and mean. One is a thin pole whose own zone reads its 2.03 m distance correctly while neighbouring zones see a wall at 2.78 m, a partial-fill failure on thin structure. The other sits outside the cone and is unsupported extrapolation, the limit of Section V-D appearing in independent data, and it accounts for the maximum error.

One caveat bounds the sensor-scored numbers above, each evaluation point there is a zone centre, precisely where nearest-neighbour assignment looks best, so error across a discontinuity falling *between* zones cannot appear. The tape reference is free of that restriction, and the thin-pole marker is this mode.

D. The far-field ceiling

Everything scored so far sits close to the sensor, since anchors and tape markers alike fall within a few metres, so nothing yet has tested the far field. The limit derived in Section IV is what governs it, and on both sensors the ceiling sits below the sensor’s own range. On our hardware it sits at 1.43 m on a typical frame, since $\hat{b}$ is refitted every frame, while the dToF reads to 4.16 m, so anchors beyond 1.5 m are 16% of the data while carrying 69% of the error, and a pooled mean conceals that because half of our anchors sit under 0.5 m. On ZJU-L5, where ground truth reaches 20 m, the consequence is a systematic under-read of -12.1 m beyond 10 m, with 885 pixels per frame, 0.3% of the valid pixels in a frame, carrying 56% of the squared error. The failure is an under-read rather than the divergence an unbounded reciprocal might suggest, which is why AbsRel and δ_1 in Table VI stay competitive while RMSE does not. The sign matters for the application, an under-read places distant structure closer than

TABLE IV
RIDGE ON THE SHIFT TERM, ZJU-L5 TRAIN SPLIT

ridge	RMSE↓	MAE↓	AbsRel↓	$\delta_1 \uparrow$	bias→ 0
0 (default)	1.012	0.261	0.105	0.886	-0.135
0.003	0.962	0.246	0.097	0.890	-0.091
0.01	1.085	0.278	0.103	0.896	-0.014
0.03	1.302	0.362	0.139	0.858	+0.105
∞ (scale-only)	3.006	1.155	0.529	0.569	+0.927

it is, the conservative direction for collision avoidance, though it remains an error for mapping.

The obvious remedy is to remove the ceiling, and Table IV shows why that is wrong. When a ridge on b is swept from the plain affine fit to exact scale-only, the scale-only endpoint triples RMSE and flips the bias positive. Scale-only has no ceiling, since $D = 1/(a\hat{\delta})$ is unbounded, but at long range the disparity is small and the reciprocal becomes extremely sensitive to noise, which b regularizes against. The ceiling is a bias-variance control rather than simply a defect, so a log-depth reparameterization or unconstrained shift is refuted rather than untried. A small ridge does help: at 0.003, selected on the train split and applied unchanged at test, AbsRel improves by 8% and RMSE by 5% with no near-field cost, while past 0.01 an unfavourable trade appears. Two further remedies are ineffective, capping depth at a multiple of the furthest anchor is monotonically worse than the fixed ceiling, since capping only reduces depth and the error is already an under-read, and using the predicted variance as a rejection filter fails for the reasons Section V-E gives.

Swept on our own logs, that same strength moves the ceiling only from 1.43 to 1.45 m, because our anchors have a median range of 0.49 m against ZJU-L5’s 1.59 m and the shift term is therefore dominated by near anchors. The ridge stays at zero in deployment. The deployed system instead mitigates how the failure is reported, marking pixels at the reciprocal singularity invalid rather than assigning them an arbitrarily large depth, and it floors the published variance at 100% relative uncertainty outside the measured region.

E. Uncertainty calibration

Flooring the variance outside the measured region only helps if the variance means something, which is the third question of Section I. At anchor pixels the published variance correlates with realized error at 0.943, but that regime is circular and near-range, so we evaluate it against the full 15-point tape reference, whose points were chosen to contain the cases that break it: four outside the dToF cone to 38.9° , three straddling its edge, and four on depth discontinuities, with σ from 0.06 to 3.3 m. Repeat captures matter here, since two taken two minutes apart on a stationary scene differ in rank correlation by 0.13 from pipeline noise alone, so no single capture should be read more finely than that.

Three further quantities drawn from arbitration feed the variance, each addressing a regime the analytic term of Equation (10) cannot see: how far arbitration displaces the depth, the angular distance to the nearest anchor, and whether a zone

TABLE V
PREDICTED UNCERTAINTY AGAINST TAPE-MEASURED ERROR

	$n=11$ tuned on	$n=15$ all points
rank corr($\sigma, e $) $\uparrow$	0.745	0.613
coverage @ 1σ ($\rightarrow$ 0.683)	0.909	0.760
coverage @ 2σ ($\rightarrow$ 0.954)	0.909	1.000
worst residual↓	2.30σ	1.71σ

disagrees with its neighbours. Table V scores them against both the set the constants were fitted on and the larger one.

The constants were tuned on eleven of the points and are scored here on all fifteen. Rank correlation falls from 0.745 to 0.613 across that step, a gap no larger than the 0.13 swing two captures of a motionless scene already produce, so the constants are not simply describing their own calibration data. Coverage moves toward its target rather than away, from an over-conservative 0.909 at 1σ to 0.760 against the nominal 0.683. A separate ablation shows what the arbitration terms buy. Without them the worst residual is 3.86σ , and with them it is 2.30σ .

The estimate is most valuable outside the cone, where the pipeline’s error reaches 1.07 and 1.52 m while it reports σ of 2.10 and 1.84, declaring that it is extrapolating. Those three terms cost 10.6 ms per frame at the deployed full resolution rather than the 480×640 Table II reports, so the 25.8 ms there excludes them while the deployed 9.4 Hz includes them. For that cost the interval is usable as a ranking and conservative as a bound. Inside the measured region its width is set by the learned term τ^2 , and outside it by the 100% relative floor of Section V-D. The analytic ceiling of Section IV bounds V_{fit} , but that term contributes roughly 0.1% of the published variance, so it is not what sets the width.

F. Comparison with published methods

Every number so far comes from our own sensor and our own camera. Table VI reports ZJU-L5, whose input is 8×8 rather than 32×32 and whose ground truth is dense RealSense from another laboratory. The analytic stage consumes that layout unmodified, which demonstrates the layout independence of Section IV-C on hardware we do not own.

The comparison is compute-fair, since our competitive rows run on the same DAV2 ViT-S weights DEPTHOR uses, leaving their 6M completion head as the only architectural difference. Dropping from the 335M teacher to that 24.8M backbone moves AbsRel from 0.083 to 0.091 with δ_1 nearly unchanged, so a large backbone is not carrying the result, and at matched compute the analytic fit beats both CFPNet and PENet on AbsRel and δ_1 . This answers the first question of Section I: metric alignment does not require a learned fusion module, because two parameters fitted in closed form already outperform two methods trained on this benchmark on typical-pixel metrics.

RMSE is the exception, at 2.2 to 2.8 times the published methods while typical-pixel metrics stay competitive. That combination is only possible if a small minority of pixels are severely wrong, and Section V-D identifies the mechanism; median absolute error alone, at 0.049 m, would have concealed

TABLE VI
ZERO-SHOT RESULTS ON ZJU-L5. THE COMPETITIVE “OURS” ROWS
RUN THE ANCHORING STAGE ALONE AND DIFFER ONLY IN BACKBONE;
THE DEPLOYED ROW IS THE COMPLETE 4.1M PIPELINE; RIDGE
STRENGTH IS SELECTED ON THE TRAIN SPLIT

Method	backbone	AbsRel↓	RMSE↓	$\delta_1 \uparrow$
CFPNet [2]	—	0.103	0.431	0.883
PENet [13]	—	0.161	0.498	0.807
Ours, affine	ViT-S 24.8M	0.094	1.056	0.902
Ours, ridge on b	ViT-S 24.8M	0.091	1.031	0.908
Ours, ridge on b	ViT-L 335M	0.083	0.953	0.913
DEPTHOR-Small [3]	ViT-S +6M	0.079	0.371	0.923
DEPTHOR-Large [3]	ViT-S +12M	0.075	0.350	0.933
<i>Ours, as deployed</i>	<i>distilled 3.66M</i>	<i>0.185</i>	<i>1.174</i>	<i>0.716</i>

Rows are limited to methods we could re-score through our own masks; DELTAR [1] and DEPTHOR++ [4] also report on ZJU-L5. Accuracy here is for the competitive rows; Table II gives embedded latency for the deployed one. Our RMSE is a far-field artefact rather than a whole-frame one, 0.418 inside the dToF footprint against 1.398 outside on the ViT-S row, for the reason Section V-D gives.

it. The tail is localized as well, with RMSE 1.398 outside the dToF footprint against 0.418 inside, where the analytic path reaches $\delta_1 = 0.968$ and AbsRel 0.052. The deficit is therefore far-field extrapolation failing where the sensor never measures, not a general loss of accuracy.

Re-scoring their model through our own metrics and masks, which reproduces their published figures to within 0.002 to 0.004, puts our deficit at 6% on AbsRel over all valid ground truth, 4% inside the footprint and 8% outside, so region restriction flatters neither method. Those figures use our own valid-pixel masks, not the whole-frame comparison of Table VI, where the same row sits 15% behind on AbsRel. That uniform 4 to 8% is what zero learned fusion costs against a 6M completion head at matched backbone. The deployed configuration then trades further benchmark accuracy for the $3.1\times$ speedup of Table II.

One caveat bounds all of the above. At the ZJU-L5 anchors the deployed student’s predicted disparity correlates with inverse ground-truth depth at $\rho = 0.417$ against the teacher’s 0.867, and under that backbone nearest-zone assignment beats the pipeline outright. Distilling on a single lens is what buys the speed and what costs the transfer, so layout independence is a property of the estimator while domain generality is a property of the backbone, and only the first is claimed here.

VI. DISCUSSION AND LIMITATIONS

Metric alignment of a monocular prior to sparse dToF measurements does not require a learned fusion module, since a two-parameter analytic fit exceeds two in-domain trained methods on relative error and threshold accuracy, though not on RMSE. Where a learned residual does help, the geometry of its supervision matters more than its capacity. A robot too small for a scanning LiDAR can therefore still publish the metric point cloud its navigation and mapping stack expects. We achieve substantial speedups, making the system able to run on embedded hardware at the expense of some accuracy. At matched backbone the fit stays within 4 to 8% of DEPTHOR-Small while removing its 6M completion head, and the deployed configuration trades further accuracy for a $3.1\times$ latency advantage on the robot’s own processor.

The clearest next step is to add temporal fusion to the anchoring stage. Surfaces that lie beyond the current ceiling enter the dToF cone as the platform moves, so accumulating anchors across frames would constrain the far field retroactively and would extend metric support into the periphery.

There are four limitations to record. First, the on-robot figures are scored against the same sensor that supervises the system, which is why Section V-C validates against an independent tape reference, though that reference stops at 4.03 m rather than the full span the affine form must express. Second, a further next step is to collect data across varied lighting in various field environments. Third, the deployed backbone does not transfer to another camera, so our efficiency and benchmark-accuracy figures describe different configurations. Fourth, too few valid returns degrade the conditioning of the fit, after which the system falls back to a scale-only estimate without signalling it, and exposing fit conditioning as a health signal would close that gap.

VII. CONCLUSION

This work establishes metric scale for a monocular depth prior from multizone dToF measurements using a closed-form two-parameter fit, confining learning to relative structure and local correction. Evaluated zero-shot on ZJU-L5 against dense independent ground truth and on the same backbone DEPTHOR uses, that fit reaches $\delta_1 = 0.902$ at AbsRel 0.094 with no retraining for the smaller zone layout, ahead of two methods trained on the benchmark and uniformly 4 to 8% behind DEPTHOR-Small, while the deployed pipeline runs $3.1\times$ faster than that method’s network and publishes at 9.4 Hz on the robot. Whether the refiner contributes at all is set by the geometry of its supervision, since a contiguous island improves median extrapolation error by 40% where a scattered hold-out improves nothing.

REFERENCES

- [1] Y. Li *et al.*, “DELTAR: Depth estimation from a light-weight ToF sensor and RGB image,” in *Proc. ECCV*, 2022.
- [2] L. Ding, H. Jiang, R. Xu, and R. Huang, “CFPNet: Improving lightweight ToF depth completion via cross-zone feature propagation,” in *Proc. 3DV*, 2025.
- [3] J. Xiang *et al.*, “DEPTHOR: Depth enhancement from a practical light-weight dToF sensor and RGB image,” in *Proc. ICCV*, 2025.
- [4] J. Xiang *et al.*, “DEPTHOR++: Robust depth enhancement from a real-world lightweight dToF and RGB guidance,” arXiv:2509.26498, 2025.
- [5] L. Ding, H. Jiang, J. Chen, and R. Huang, “Self-supervised enhancement for depth from a lightweight ToF sensor with monocular images,” in *Proc. IROS*, 2025.
- [6] H. Lin *et al.*, “Prompting Depth Anything for 4K-resolution metric depth,” in *Proc. CVPR*, 2025.
- [7] J. Park, K. Joo, Z. Hu, C.-K. Liu, and I. S. Kweon, “Non-local spatial propagation network for depth completion,” in *Proc. ECCV*, 2020.
- [8] Y. Zhang *et al.*, “CompletionFormer: Depth completion with convolutions and vision transformers,” in *Proc. CVPR*, 2023.
- [9] L. Yang *et al.*, “Depth Anything V2,” in *Proc. NeurIPS*, 2024.
- [10] M. Hu *et al.*, “Metric3D v2: A versatile monocular geometric foundation model,” *IEEE Trans. Pattern Anal. Mach. Intell.*, 2024.
- [11] L. Piccinelli *et al.*, “UniDepth: Universal monocular metric depth,” in *Proc. CVPR*, 2024.
- [12] D. Wofk, R. Ranftl, M. Müller, and V. Koltun, “Monocular visual-inertial depth estimation,” in *Proc. ICRA*, 2023.
- [13] M. Hu *et al.*, “PENet: Towards precise and efficient image guided depth completion,” in *Proc. ICRA*, 2021.